Heritage Archive Explorer

Multimodal Retrieval-Augmented Generation for Cultural Heritage Archives

Generative AI | Digital Libraries | UNESCO Alignment

Team CodeNova | CSGS Hackathon Spring 2026
Old Dominion University

The Archive Access Gap – The Problem

882,000+ manuscripts, maps and photographs digitized globally - yet invisible to the public

Keyword search requires expert knowledge - students, tourists and researchers are left out

UNESCO's digitization mission is growing, but the accessibility gap is widening

"Digitized but not Democratized"

Heritage Archive Explorer – Our Solution

Natural Language Querying

Ask anything in plain English or French

Multimodal Retrieval

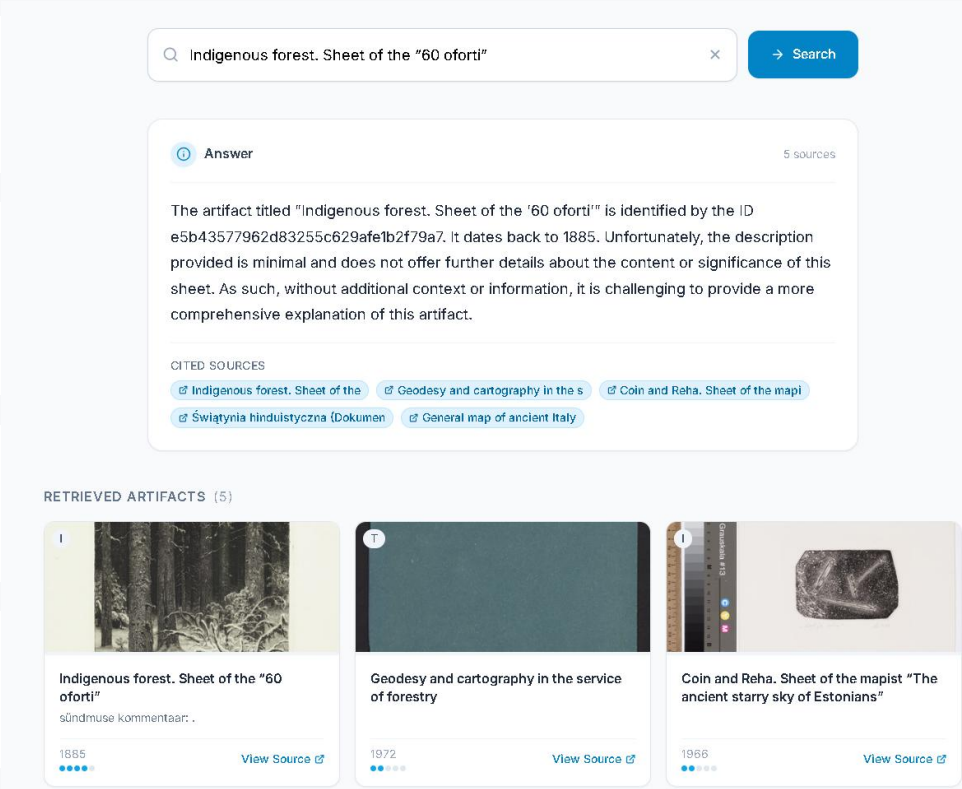
Manuscripts, maps, photographs and texts - all searchable

Grounded AI Answers

Every response cites real artifacts - no fabrication

Interactive Knowledge Graph

Visualize 882 artifacts before you even search



The RAG Pipeline

DATA PIPELINE

Europeana API

↓

882 Artifacts Collected

↓

OpenAI Embeddings (1,536-dim vectors)

↓

Local Vector Store + Knowledge Graph

QUERY PIPELINE

User Question (EN or FR)

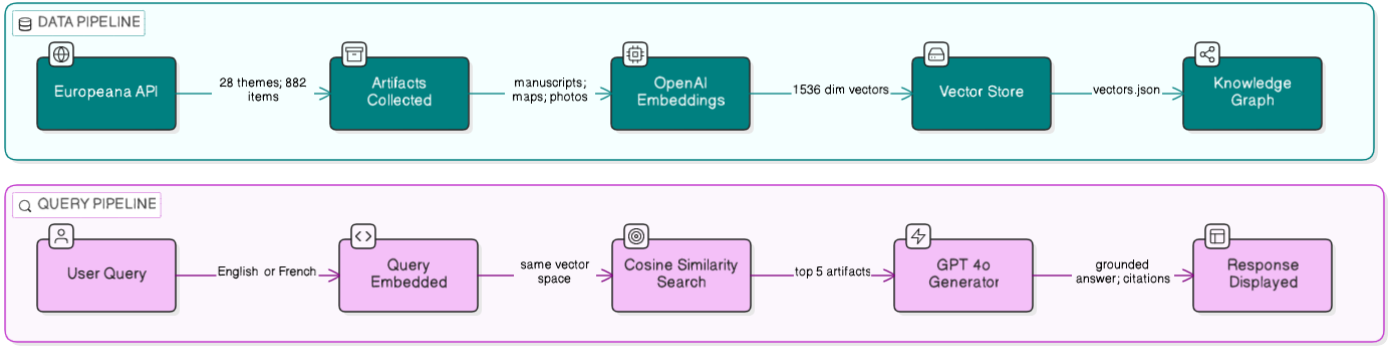
Query Embedded

Cosine Similarity Search

Top 5 Artifacts Retrieved

GPT-4o Generates Cited Answer

Answer + Artifact Cards Displayed



RAG Benchmark Evaluation

Tested on 10 curated queries across both languages and all artifact types

CONTEXT RECALL

1.00

Every query found relevant artifacts

SOURCE COVERAGE

1.00

All retrieval slots filled every time

FAITHFULNESS

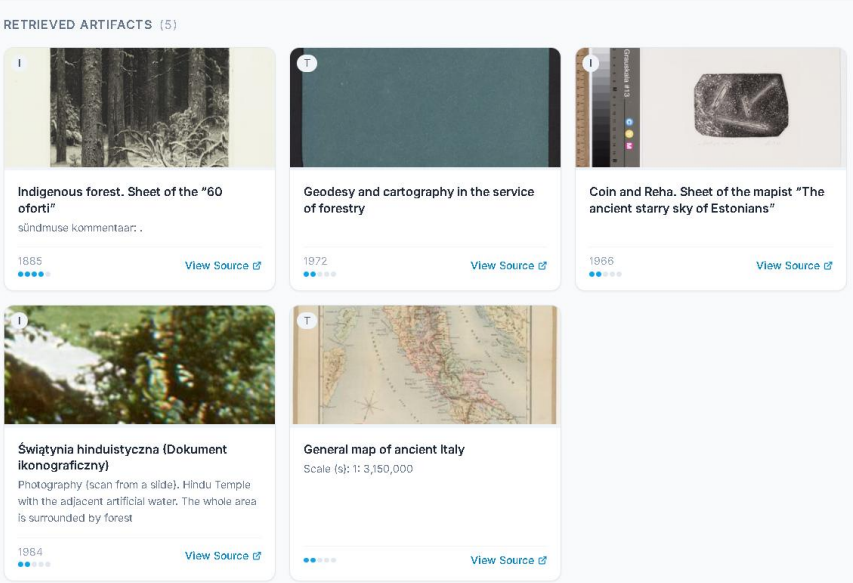
0.70

Answers grounded in retrieved sources

ANSWER RELEVANCY

0.60

Responses address the question aske



Overall RAG Score: 0.82 / 1.00

"Retrieval is perfect. Faithfulness is strong. Relevancy improvement identified as future work via query decomposition."

Real World Impact

Who It Serves

- Students & Researchers
- Tourists & Journalists
- Educators & Museums

What It Enables

- Multilingual access to world heritage
- Transparent, cited AI answers
- No expertise required to explore archives

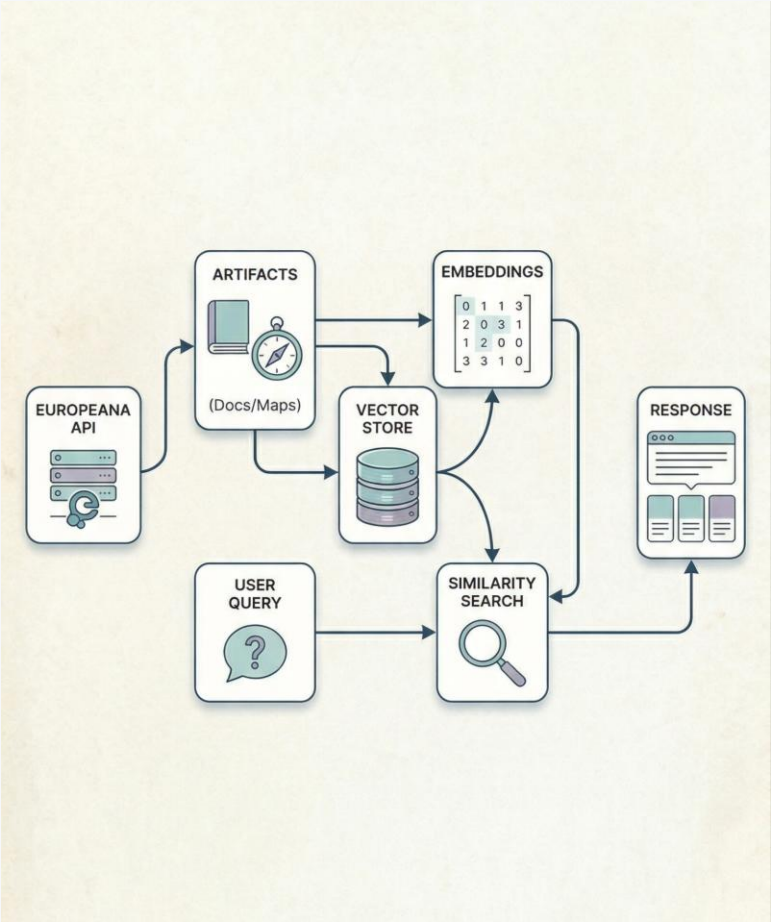
Where It Scales

- Any Europeana collection
- Any language GPT-4o supports
- Deployable for any digitized archive globally

"We didn't build a search engine. We built a guide to human history."

System Architecture Overview

Layer	Technology
Data Source	Europeana Open API
Embeddings	OpenAI text-embedding-3-small (1,536-dim)
Vector Store	ChromaDB – cosine similarity search
Generation Model	GPT-4o – only grounding prompt
Backend	FastAPI Python
Frontend	React, Vite, Tailwind CSS, D3.js
Languages Supported	English & French



See It in Action

Ask "Show me maps of the Ottoman Empire"

Instant results with citations

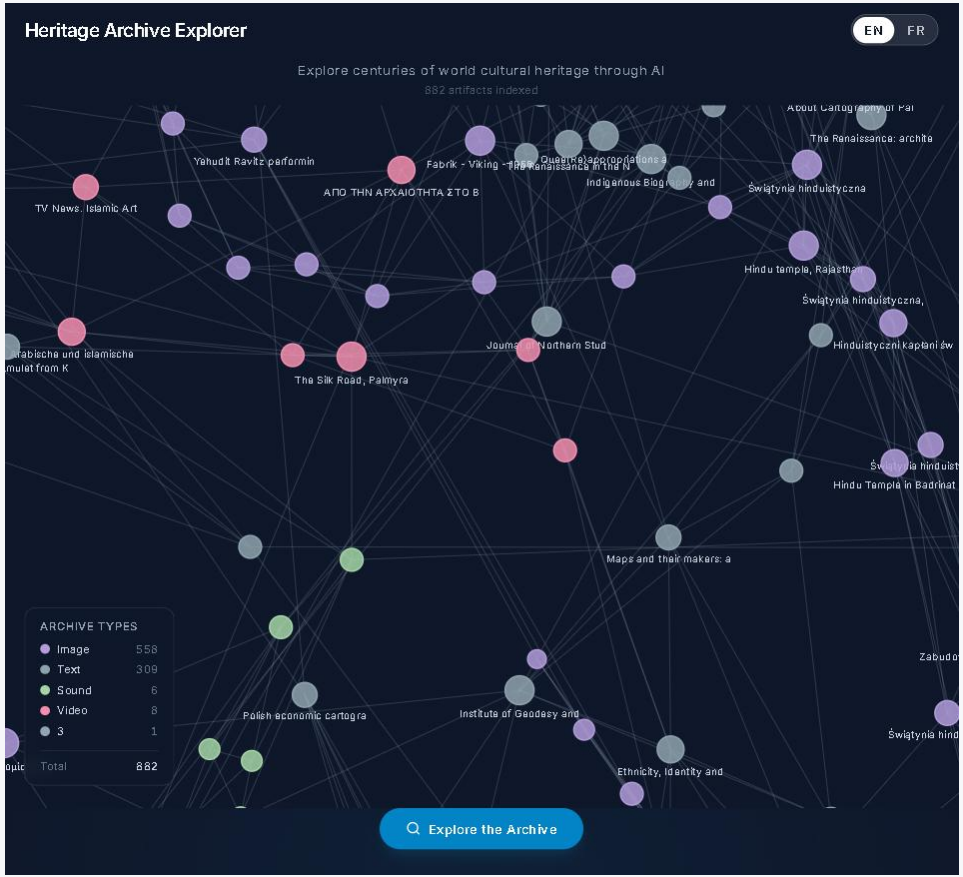
Switch language to French

Seamless multilingual retrieval

Explore the Knowledge Graph

882 artifacts visualized as a network

Click any artifact

Full metadata, image, and source link

No login. No expertise. Just ask.